

Erdős Problem #761

EP-761	L1 - Tractable	Partiellement résolu	Graph Theory
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Énoncé original (anglais, reproduction mot pour mot)

The cochromatic number of G , denoted by $\zeta(G)$, is the minimum number of colours needed to colour the vertices of G such that each colour class induces either a complete graph or empty graph. The dichromatic number of G , denoted by $\delta(G)$, is the minimum number k of colours required such that, in any orientation of the edges of G , there is a k -colouring of the vertices of G such that there are no monochromatic oriented cycles.

Must a graph with large chromatic number have large dichromatic number? Must a graph with large cochromatic number contain a graph with large dichromatic number?

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